

Practical-5

Question

1. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.
2. Develop a program that executes the equivalent of the shell command **ls -l | grep ".c"** using **fork()**, **pipe()**, **dup2()**, and **exec()** system calls.

What We Did

- For the first part, we created an anonymous pipe and used **fork()** to create a parent and child process. The parent acted as the producer and sent 10, 20, 30, 40 and 50 through the pipe. The child acted as the consumer and read the same values.
- We used **clock()** to measure the time taken during the communication.
- For the second part, we recreated the behavior of **ls -l | grep ".c"**. The parent executes **ls -l** and sends its output into the pipe, while the child takes the pipe input and executes **grep ".c"**.
- **dup2()** connects the pipe to standard output/input: file descriptor **1** is used for output and **0** for input.

Commands Used

Producer-consumer: gcc Producer.c → ./a.out

Pipe + exec: gcc Practical-5_exec.c → ./a.out

Output

The producer-consumer program successfully produced and consumed all five values. The measured time was displayed in the terminal. The pipe + exec program successfully displayed the .c files from the directory, matching the behavior of **ls -l | grep ".c"**.

Pipe + exec Output

```
manu@Manasvi:~/os_lab$ gedit Practical-5_exec.c
** (gedit:4971): WARNING **: 03:31:32.332: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:4971): WARNING **: 03:31:32.332: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
manu@Manasvi:~/os_lab$ gcc Practical-5_exec.c
manu@Manasvi:~/os_lab$ ./a.out
-rw-r--r-- 1 manu manu 293 Aug 24 18:08 Practical-2.c
-rw-r--r-- 1 manu manu 382 Aug 24 18:32 Practical-3.c
-rw-r--r-- 1 manu manu 413 Sep 24 03:31 Practical-5_exec.c
-rw-r--r-- 1 manu manu 302 Aug 24 17:07 Practical_1.c
-rw-r--r-- 1 manu manu 537 Aug 30 19:45 forkexec.c
manu@Manasvi:~/os_lab$
```

Producer-Consumer Output

```
manu@Manasvi:~$ gcc Producer.c
manu@Manasvi:~$ ./a.out
producer
produced 10
produced 20
produced 30
produced 40
produced 50
time taken is 0.000133
consumer
Consumed 10
Consumed 20
Consumed 30
Consumed 40
Consumed 50
time taken is 0.000585
manu@Manasvi:~$
```

```
manu@Manasvi:~$ gedit Producer.c
** (gedit:2277): WARNING **: 03:04:52.791: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2277): WARNING **: 03:04:52.791: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
manu@Manasvi:~$ gcc Producer.c
manu@Manasvi:~$ ./a.out
producer
produced 10
produced 20
produced 30
produced 40
produced 50
time taken is 0.000124
consumer
Consumed 10
Consumed 20
Consumed 30
Consumed 40
Consumed 50
time taken is 0.000163
manu@Manasvi:~$ |
```